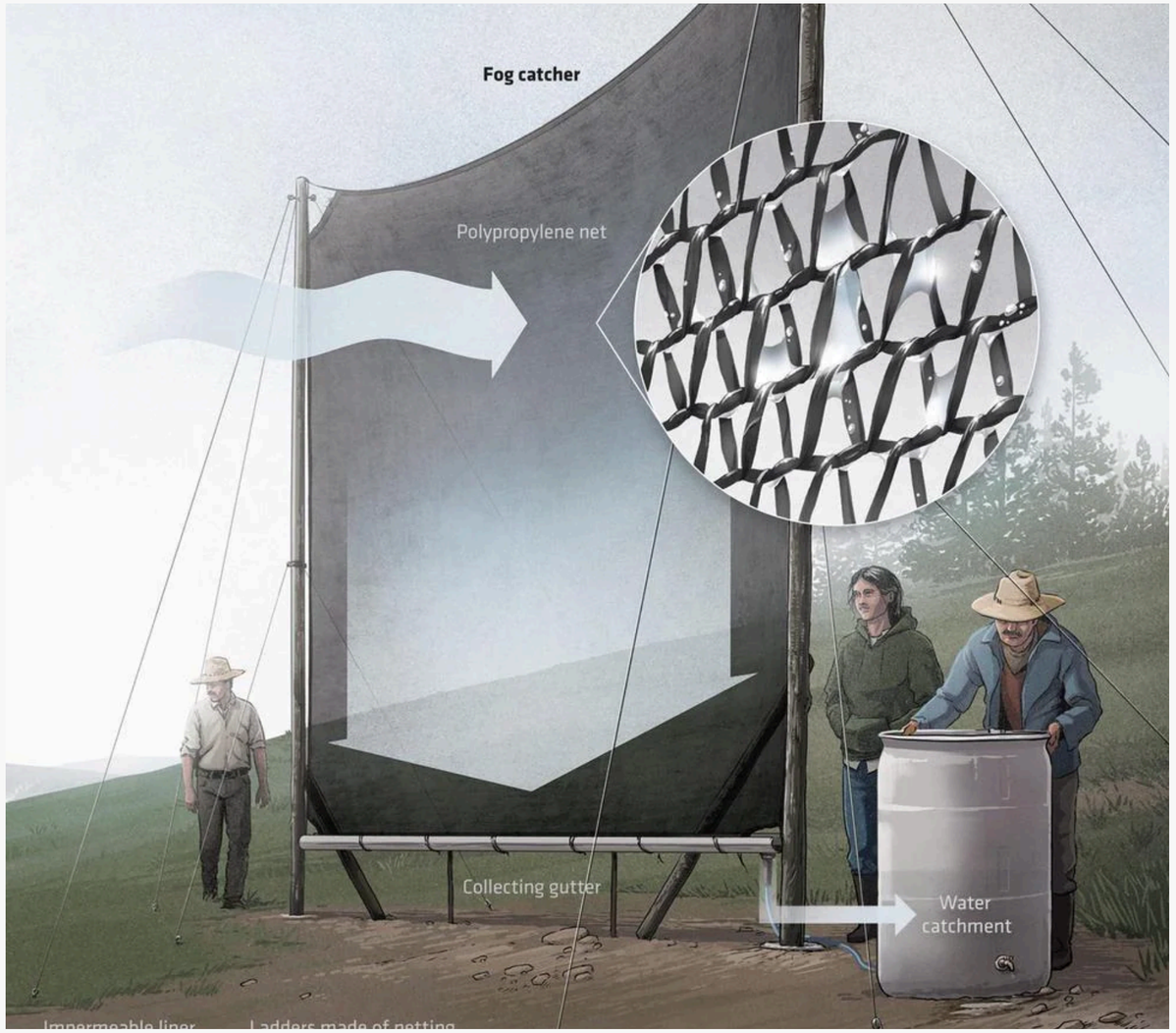
3D Fog Harvester and Simulation

Madison Dewrell

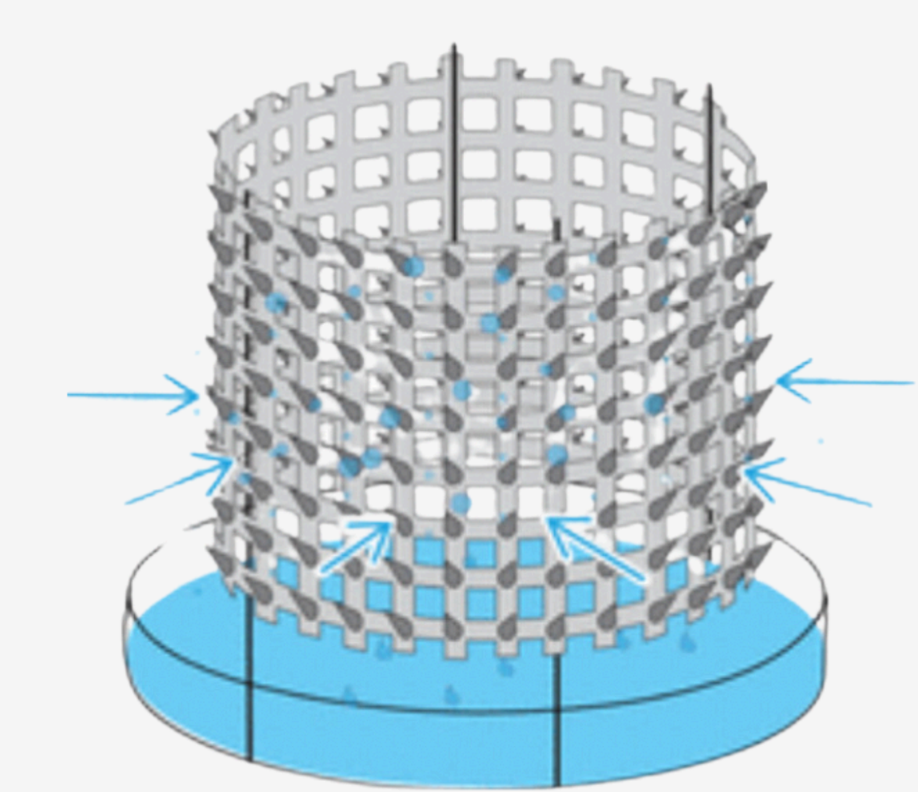
Research Project 2023-2024

What is Fog Harvesting?

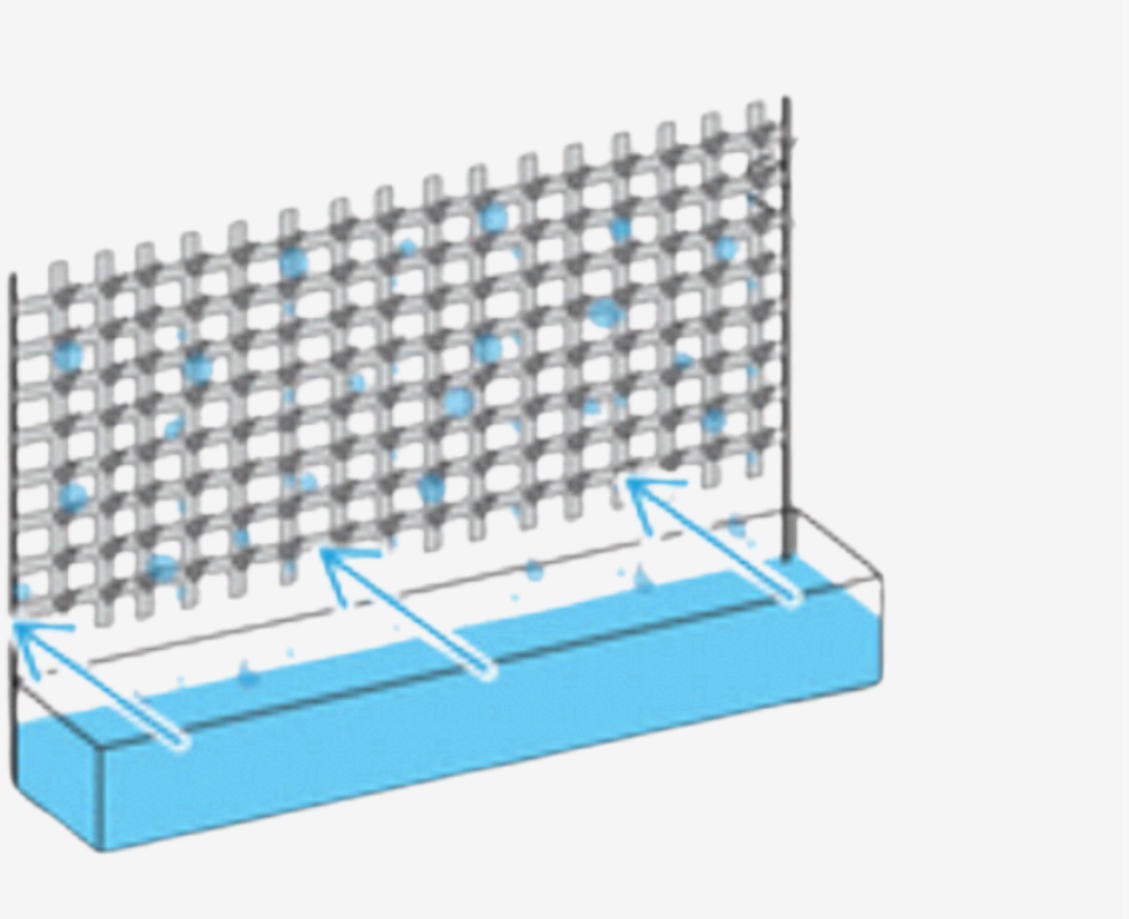




Symmetrical Vs Conventional

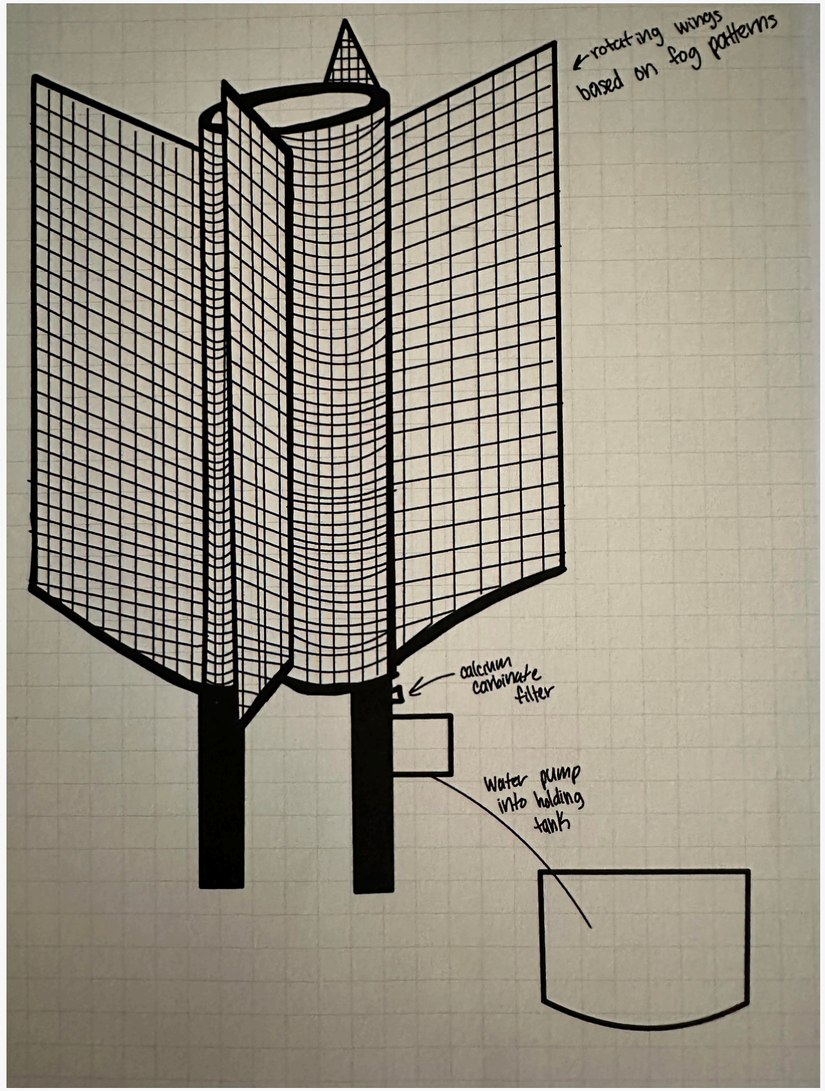


3D Net Fog Harvester



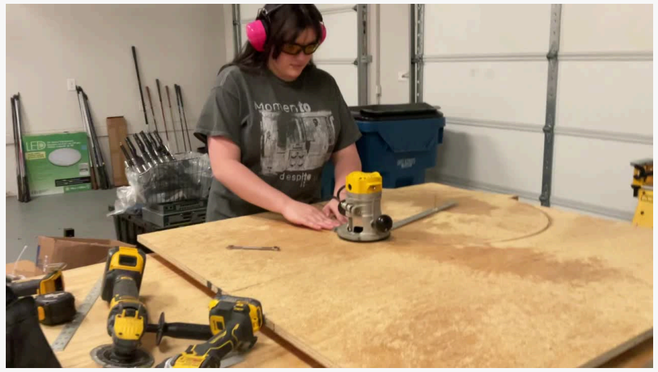
2D Net Fog Harvester

The Symmetrical Fog Harvester



Progress Board



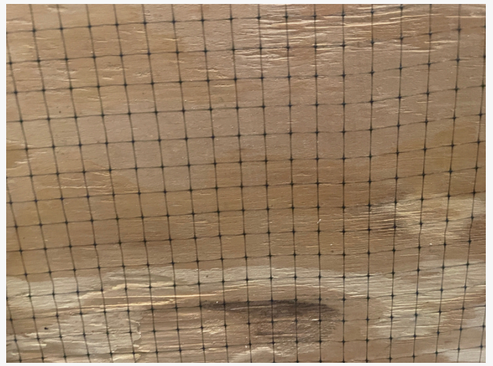


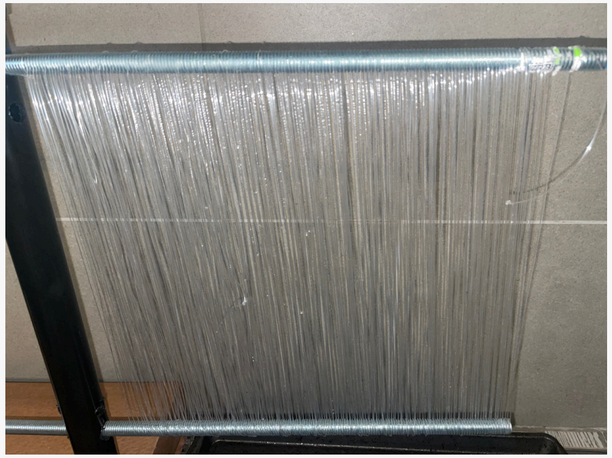












The Symmetrical Fog Harvester





Thesis

How can a Java-based climate simulation model utilizing environmental variables predict fog collection yields for different types of fog harvesters, enabling optimized designs and harvester selection tailored to diverse climates?

Simulation

$$Fw = fc \times (FPI) \times f(H) \times f(W) \times Ac$$

Fw - Volume of water collected

Fc - CONSTANT (Calibration parameter)

FPI - Fog Potential Index dependent on difference between dew point and air temperature

f(H) - Humidity function

f(W) - Wind Speed Function

Ac - Surface Area

Simulation

1aa. $FPI = (T_d + 1.0 - T) \times 2.6;$
$for (T_d + 1.0) > T = 0;$
$for (T_d + 1.0) \leq T$

T_d = Dew Point
T = Dry bulb temperature

1ab. $T_d = 243.5 \times (\ln (e/6.112)) / (17.67 - \ln (e/6.112))$

where 'e' is the actual vapor pressure in millibar, which can be calculated using following equation:

$RH = 100 \times (e/e_S)$
E_s calculation:

$e_S = 6.112 \times \exp (17.67 \times T / T + 243.5)$

1b. Humidity Function
$f(H)_i = (RH_i + RH_{i-1})/2 \times 100$

$f(W) = 0.4 \times V;$
for $V < 4$ m/s $= 3/V;$
for $V \geq 4$ m/s

V = Wind Velocity

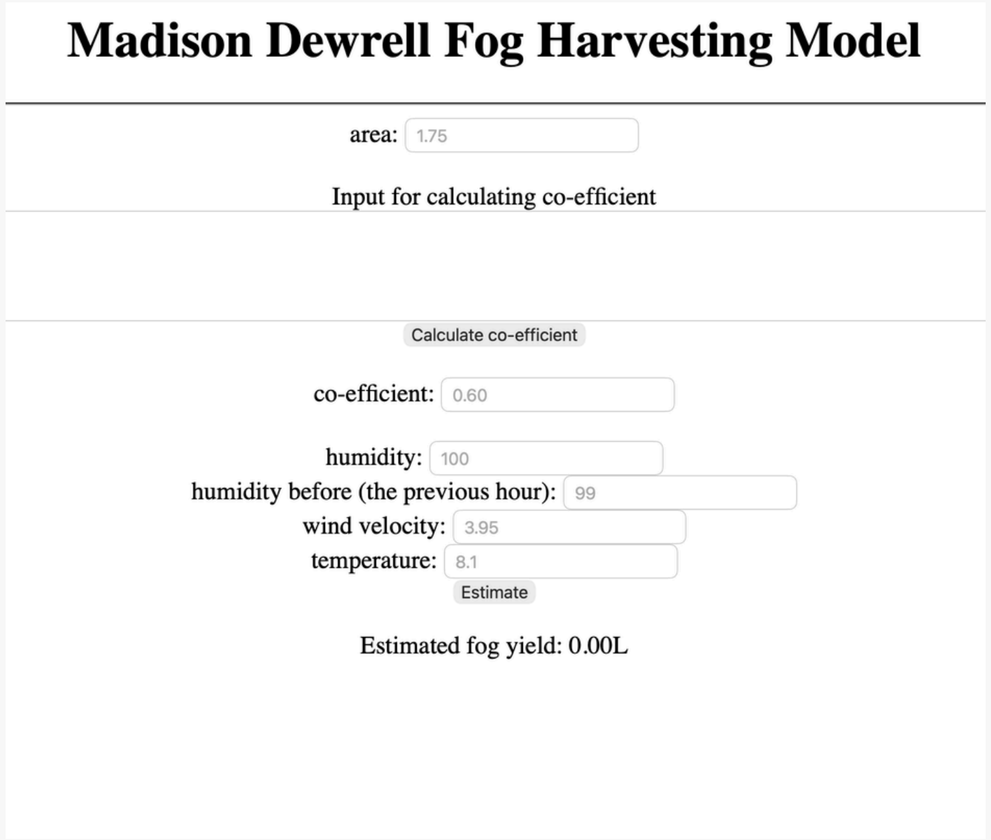
1c. Wind Speed Function
$f(W) = 0.4 \times V;$ for $V < 4$ m/s $= 3/V;$ for $V \geq 4$ m/s

Simulation Code

Simulation Code

```
    <header class="text-center">
        <h1 class="text-2xl font-bold">Fog Harvesting Model MTD</h1>
        <hr>
    </header>
    <main>
        <label for="area">area (m²):</label>
        <input type="text" id="area" name="area" placeholder="1.00"></input>
        <br>
        <br>
        <label for="calculate">Input for calculating co-efficient</label>
        <br>
        <textarea name="calculate" id="calculate" style="width: 100%" rows="5"
        placeholder="relative humidity (%), wind speed (m/s), Air temperature (°C), Fog yield (L)
100,7.5,6.95,0
100,10.10,6.5,0.83
100,12.75,6.15,0.82"></textarea>
        <br>
        <button onclick="calculate()">Calculate co-efficient</button>
        <br>
        <br>
        <label for="co-efficient">co-efficient:</label>
        <input type="text" id="co-efficient" name="co-efficient" placeholder="1.01"></input>
        <br>
        <br>
        <label for="humidity">humidity (%):</label>
        <input type="text" id="humidity" name="humidity" placeholder="100"></input>
        <br>
        <label for="humidity-before">humidity before (the previous hour) (%):</label>
        <input type="text" id="humidity-before" name="humidity-before" placeholder="99"></input>
        <br>
        <label for="wind-velocity">wind velocity (m/s):</label>
        <input type="text" id="wind-velocity" name="wind-velocity" placeholder="3.95"></input>
        <br>
        <label for="temperature">temperature (°C):</label>
        <input type="text" id="temperature" name="temperature" placeholder="8.1"></input>
        <br>
        <button onclick="estimate()">Estimate</button>
        <br>
        <p>Estimated fog yield: <span id="fog-yield">0.00</span>L</p>
    </main>
```

Simulation Code





Testing



Model Predictions

Location One: Kennesaw Mountain

Hours	Relative Humidity (%)	Wind Speed (m/s)	Air Temperature (C)	Expected Fog Yield (L)	Actual Fog Yield (L)
1	100	7.5	31.2	0.00	0
2	100	10.1	31.0	0.00	0
3	100	12.75	32.7	0.00	0
4	99	15.5	32.9	0.00	0
5	98	11.9	36.0	0.00	0
6	100	9.2	36.4	0.00	0

Hours	Relative Humidity (%)	Wind Speed (m/s)	Air Temperature (C)	Expected Fog Yield (L)	Actual Fog Yield (L)
1	87	5.6	31.2	0.000	0.2
2	99	13.6	31.0	0.770	0.53
3	99	11.0	32.7	1.011	0.52
4	99	12.1	32.9	0.919	0.65
5	90	5.2	36.0	0.000	0.45
6	83	14.6	36.4	0.000	0.35

Location Two: Mount Oglethorpe

Hours	Relative Humidity (%)	Wind Speed (m/s)	Air Temperature (C)	Expected Fog Yield (L)	Actual Fog Yield (L)
1	85	6.2	30.8	0.00	0
2	92	8.7	31.3	0.00	0
3	88	11.4	33.0	0.00	0
4	90	9.9	34.2	0.00	0
5	87	10.5	35.7	0.00	0
6	91	8.2	36.3	0.00	0

Hours	Relative Humidity (%)	Wind Speed (m/s)	Air Temperature (C)	Expected Fog Yield (L)	Actual Fog Yield (L)
1	89	7.0	31.5	0.00	0
2	93	8.9	32.8	0.00	0
3	86	10.7	33.4	0.00	0
4	84	13.0	33.4	0.00	0
5	92	9.4	35.9	0.00	0
6	88	7.4	36.7	0.00	0

Findings

Location One: Kennesaw Mountain

Hours	Relative Humidity (%)	Wind Speed (m/s)	Air Temperature (C)	Fog Yield (L)
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3	100	12.75	32.7	0
4	99	15.5	32.9	0
5	98	11.9	36.0	0
6	100	9.2	36.4	0

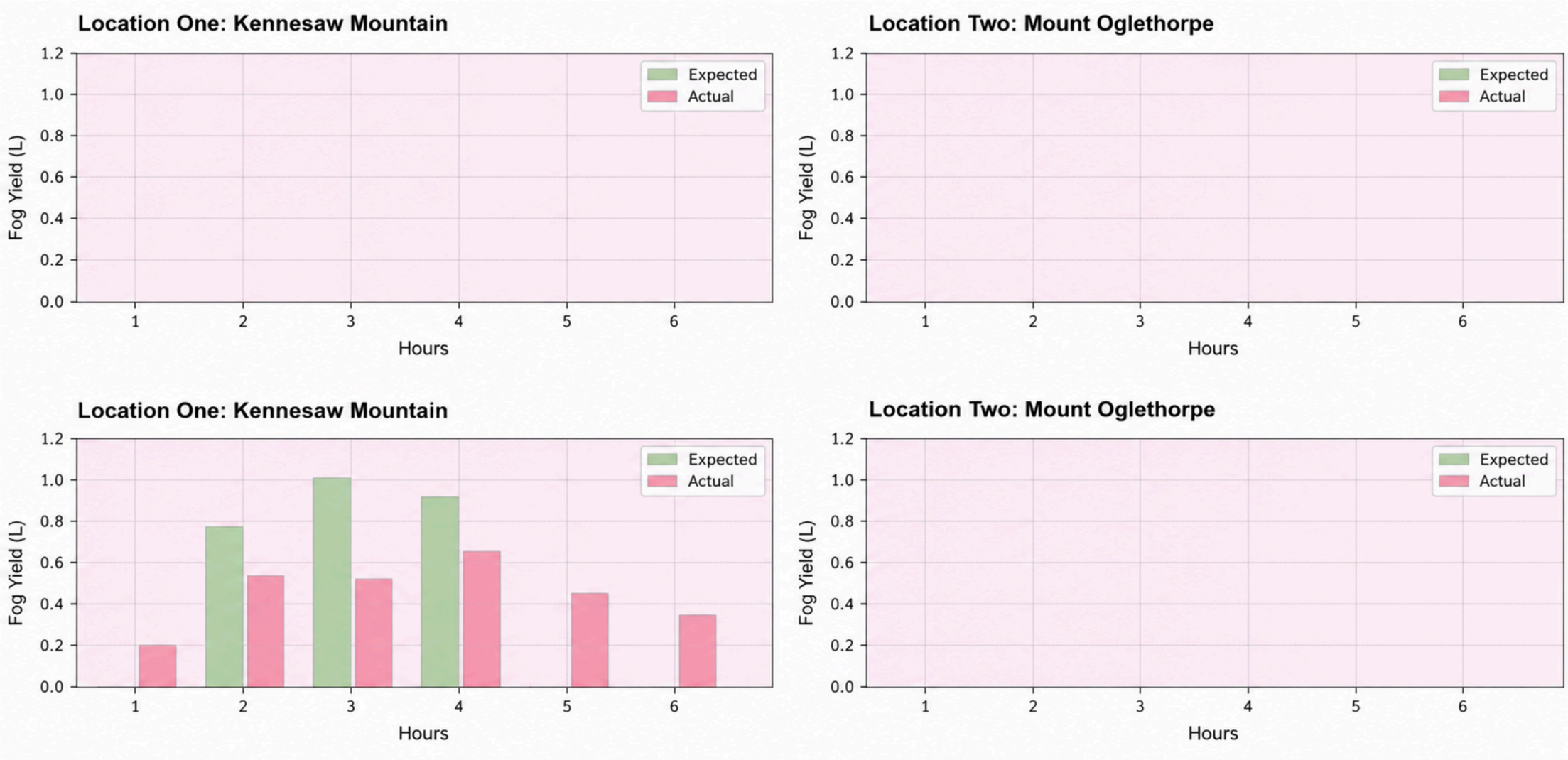
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4	84	13.0	33.4	0
5	92	9.4	35.9	0
6	88	7.4	36.7	0

Model Vs Experimental Data



What I Would Do Differently

To improve the accuracy of the fog harvesting model, I would collect more data over a longer period of time, and under a variety of weather conditions. I would also adjust the model's equations to better account for changes in humidity, wind speed, and temperature, since our current model sometimes predicts zero fog yield when water is actually collected.

Summery

The fog harvesting research project explored how fog can provide an alternative source of water through the development of a symmetrical fog harvester and a climate simulation model. By testing at Kennesaw Mountain and Mount Oglethorpe, I compared the predicted fog yields with actual collected water. The results showed that while the model could estimate fog collection under certain conditions, it was not always accurate. Overall, this project demonstrated the potential of fog harvesting and the importance of improving the model to better predict water collection in different environments.